

A CHARACTERIZATION OF A NORMAL INTRAClass COVARIANCE MATRIX

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SUMMARY

Let $X[n \times 1]$ and $Y[m \times 1]$ be normally distributed with positive definite covariance matrix Σ ; let $\bar{X} = \sum_{i=1}^n X_i/n$, $\bar{Y} = \sum_{j=1}^m Y_j/m$, $S = \sum_{i=1}^n (X_i - \bar{X})^2$, $V = \sum_{j=1}^m (Y_j - \bar{Y})^2$. Then S and V are independent, (S, V) is independent of $(\bar{X}, \bar{Y})$ and (for positive a, b) S/a and V/b have chisquare distributions if and only if Σ has a certain "intraclass" structure.

1. Introduction

The traditional statistic used in inferences about the mean of a univariate normal distribution is $T = (\bar{X} - \mu)/\sqrt{S/n(n-1)}$ where $X_1, X_2, \dots, X_n$ is the "sample", $\bar{X} = \sum_{i=1}^n X_i/n$, $S = \sum_{i=1}^n (X_i - \bar{X})^2$. However, when X_i, X_j have covariance $\sigma^2(1 + (\delta_{ij} - 1)(1 - \rho))$, it is $T_1 = T\sqrt{(1 - \rho)/(1 + (n-1)\rho)}$ which has the Student distribution. Then, $P(|T_1| > c) = \epsilon$, implies

$$P(|T| > c) = P(|T_1| > c\sqrt{(1 - \rho)/(1 + (n-1)\rho)})$$

is less than or greater than ϵ according as ρ is negative or positive.

In other words, if a "sample" is actually "correlated", confidence coefficients (mutatis mutandis significance levels) may be entirely false when the variable T is considered to have a Student distribution. Walsh (1947) considered these with some numerical detail along with corresponding results for one variance, the ratio of two variances and the difference of two means. The purpose of this note is to show that the conditions imposed essentially characterize the covariance matrix.

*Credit is due to a referee whose comments improved the paper. In particular, the scalar $c = i'Z_j$ reduces KZL to e_{ij} , a proper parameterization.

2. Preliminaries

Let $i[j]$ be an $n \times 1[m \times 1]$ vector of all ones, $I[J]$ be an $n \times n[m \times m]$ identity matrix, $K[L]$ be an $n \times n[m \times m]$ matrix of all ones. $P = I - K/n = P' = P^2 = P^+$ and $Q = I - L/m = Q' = Q^2 = Q^+$ where $'$ denotes transpose and $^+$ denotes generalized inverse. $K^+ = K/n$; $L^+ = L/m$. (Rao and Mitra, (1971) Chapter 5).

Let Δ be an $n \times n$ orthogonal matrix with all elements in the first row equal $1/\sqrt{n}$. Then $D_1 = \Delta(aI + (c-a)K/n)\Delta'$ is diagonal with elements $c, a, \dots, a$. Similarly, for ∇ $m \times m$ orthogonal with first row all $1/\sqrt{m}$, $D_2 = \nabla(bJ + (a-b)L/m)\nabla'$ is diagonal with elements $d, b, \dots, b$ (Press, (1972) Chapter 2). Note that $K\Delta' = \sqrt{n}(i, \Theta_n)$ where Θ_n is an $n \times (n-1)$ matrix of all zeroes; $L\nabla' = \sqrt{m}(j, \Theta_m)$. Also $E = \Delta K Z L \nabla' = \sqrt{nm} i' Z j \begin{bmatrix} 1 \\ \cdot \\ \cdot \end{bmatrix}$ where the dots indicate appropriate matrices of zeroes.

If Y is a p -dimensional normal random variable with positive definite covariance matrix Σ , then $Y'AY$ and $Y'BY$ are independent iff (if and only if) $A\Sigma B = 0$ (Rao and Mitra, (1971) p. 178). $Y'AY$ is distributed as (noncentral) chisquare iff $A\Sigma A = A$ (Rao and Mitra, (1971) p. 171).

3. Theorem

Let $X[n \times 1]$ and $Y[m \times 1]$ be jointly distributed normal random variables with positive definite covariance matrix Σ partitioned correspondingly as $\begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$. Let $\bar{X} = X'i/n$, $\bar{Y} = Y'j/m$, $S = X'PX$, $V = Y'QY$. Then S and V are independent, (S, V) is independent of $(\bar{X}, \bar{Y})$ and S/a , V/b have (noncentral) chisquare distributions iff $\Sigma_{11} = aI + (c-a)K/n$, $\Sigma_{22} = bJ + (d-b)L/m$, $\Sigma_{12} = KZL/mn$ with a, b, c, d , positive,

$$cd > (i'Zj)^2/mn \text{ (but } Z \text{ otherwise arbitrary).}$$

Proof:

$\bar{X}$ and S are independent iff $\bar{X}^2 = X'KX/n^2$ and S are independent whence iff $P\Sigma_{11}K = 0$. S/a is distributed as chisquare iff $P\Sigma_{11}P = aP$. Then, $P\Sigma_{11}(I-K/n) = P\Sigma_{11} = aP$ so $\Sigma_{11} = aP + (I-P)W = aP + KW/n$ where W is "arbitrary" (Rao and Mitra, (1971) p. 24). Now symmetry of Σ_{11} forces $KW = W'K$ whence all columns of W have the same sum, say c , and $KW = cK$. Then $\Sigma_{11} = aP + cK/n = aI + (c-a)K/n$ with eigenvalues c (multiplicity 1) and a (multiplicity $n-1$); these must be positive when Σ_{11} is positive definite. Similarly, $\Sigma_{22} = bJ + (d-b)L/m$ for positive b and d .

For Θ an $n \times m$ matrix of all zeroes, $S = (X', Y') \begin{bmatrix} P & \Theta \\ \Theta' & \Theta' \Theta \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$ and

$$V = (X', Y') \begin{bmatrix} \Theta \Theta' & \Theta \\ \Theta' & Q \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} \text{ so } S \text{ and } V \text{ are independent iff } \begin{bmatrix} P & \Theta \\ \Theta' & \Theta' \Theta \end{bmatrix} \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} \Theta \Theta' & \Theta \\ \Theta' & Q \end{bmatrix} = 0$$

whence $P\Sigma_{12}Q = 0$. Similarly V and $\bar{X}^2$ are independent iff $Q\Sigma_{21}K = 0$; S and $\bar{Y}^2$ are independent iff $P\Sigma_{12}L = 0$. The first of these conditions yields $\Sigma_{12} = R - PRQ$; then $K\Sigma_{12}Q = 0$ implies $0 = KRQ - KPRQQ = KRQ$ so $R = U - KUQ/n$ and

$$\Sigma_{12} = U - KUQ/n - P(U - KUQ/n)Q = U - KUQ/n - PUQ = U - UQ = UL/m.$$

Then $P\Sigma_{12}L = 0$ implies $PUL = 0$ so $U = Z - PZL/m$ and

$$\Sigma_{12} = (Z - PZL/m)L/m = ZL/m - PZL/m = KZL/mn \text{ with } Z \text{ "arbitrary"}.$$

All of this proscribes the covariance matrix to be of the form

$$\Sigma = \begin{bmatrix} aI + (c-a)K/n & KZL/mn \\ LZ'K/mn & bJ + (d-b)/m \end{bmatrix}$$

The matrix $\Phi = \begin{bmatrix} \Delta & \Theta \\ \Theta' & \nabla \end{bmatrix}$ is orthogonal and $\Phi \Sigma \Phi' = \begin{bmatrix} D_1 & E/mn \\ E'/mn & D_2 \end{bmatrix}$

via results of section 2. Then the characteristic equation for Σ is

$$(a-\lambda)^{n-1} ((c-\lambda)(d-\lambda) - (i'Zj)^2/mn)(b-\lambda)^{m-1} = 0.$$

The roots $\lambda = a, b$ are already positive; the remaining roots

$$((c+d) \pm \sqrt{(c-d)^2 + 4(i'Zj)^2/mn})/2$$

will be positive iff $cd > (i'Zj)^2/mn$ as stated in the conclusion.

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